

Relevance of protein fermentation to gut health.

It is generally accepted that carbohydrate fermentation results in beneficial effects for the host because of the generation of short chain fatty acids, whereas protein fermentation is considered detrimental for the host's health. Protein fermentation mainly occurs in the distal colon, when carbohydrates get depleted and results in the production of potentially toxic metabolites such as ammonia, amines, phenols and sulfides. However, the effectivity of these metabolites has been established mainly in in vitro studies. In addition, some important bowel diseases such as colorectal cancer (CRC) and ulcerative colitis appear most often in the distal colon, which is the primary site of protein fermentation. Finally, epidemiological studies revealed that diets rich in meat are associated with the prevalence of CRC, as is the case in Western society. Importantly, meat intake not only increases fermentation of proteins but also induces increased intake of fat, heme and heterocyclic amines, which may also play a role in the development of CRC. Despite these indications, the relationship between gut health and protein fermentation has not been thoroughly investigated. In this review, the existing evidence about the potential toxicity of protein fermentation from in vitro animal and human studies will be summarized.

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